CROWDSTRIKE

Installation Guide

Worker nodes

# Kubernetes Installation Guide

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# Requirements

## Compatibility

The Helm chart can be deployed on these Kubernetes distributions:

- Amazon Elastic Kubernetes Service (EKS)
- Azure Kubernetes Service (AKS) Note: Linux nodes only
- Google Kubernetes Engine (GKE)
- Oracle Kubernetes Engine (GKE)
- Rancher K3s
- Red Hat Openshift (recommended installation with operator)

For sensor deployments on Red Hat OpenShift Container Platform 4.6 and later, see Falcon Operator.

## Network Considerations

Depending on your network environment, you might need to allow TLS traffic on port 443 between your network and our cloud's network addresses.

If your network only allows traffic by destination IP address instead of FQDN, allow TLS traffic on port 443 over the static IP addresses. For more info, see Cloud IP Addresses and FQDNs.

### CrowdStrike cloud US-1 domains

ts01-b.cloudsink.net
lfodown01-b.cloudsink.net
lfoup01-b.cloudsink.net
https://falcon.crowdstrike.com
https://assets.falcon.crowdstrike.com
https://assets-public.falcon.crowdstrike.com
https://api.crowdstrike.com
https://firehose.crowdstrike.com

## CrowdStrike cloud US-2 domains

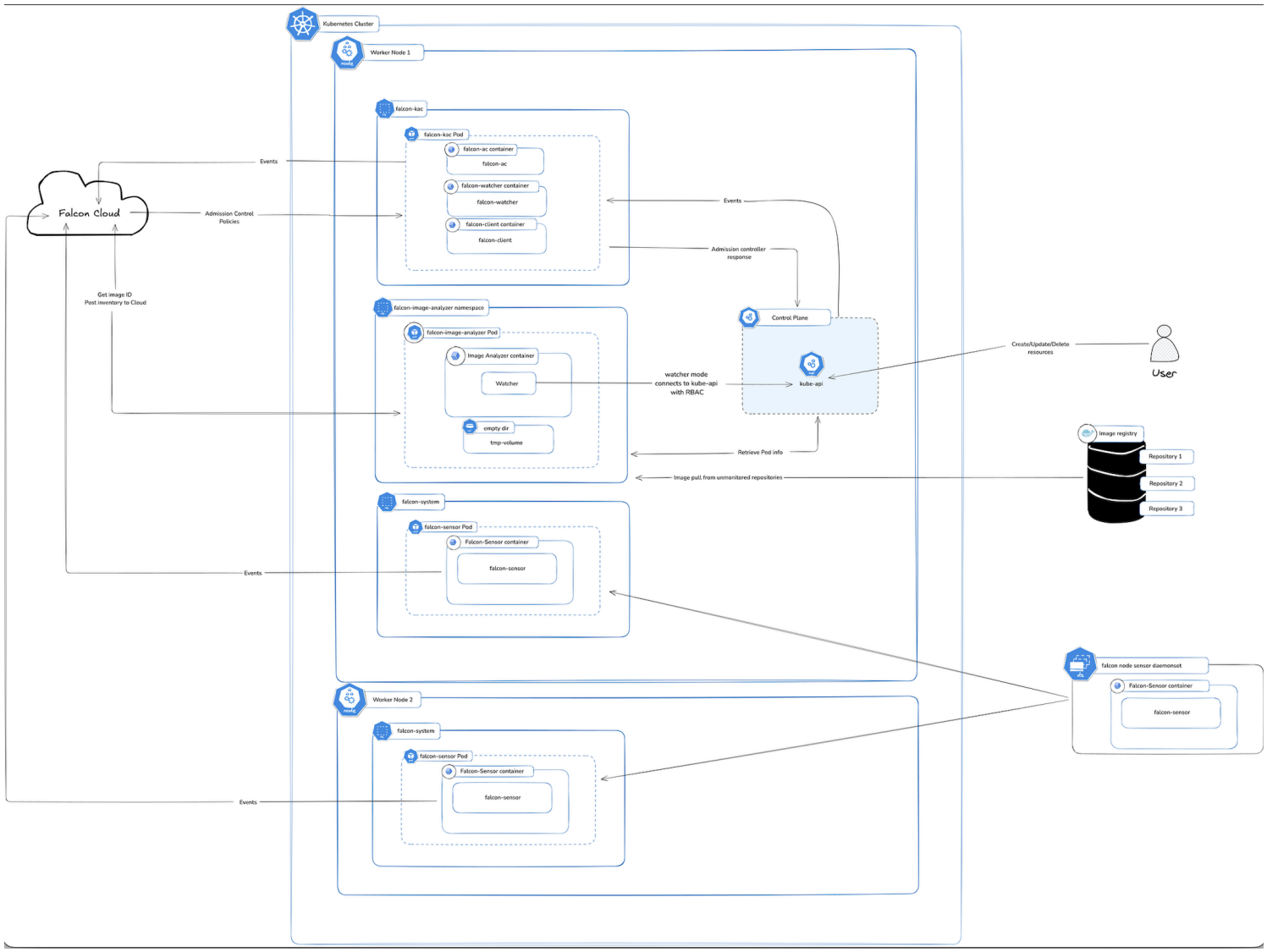
ts01-gyr-maverick.cloudsink.net
lfodown01-gyr-maverick.cloudsink.net
lfoup01-gyr-maverick.cloudsink.net
https://falcon.us-2.crowdstrike.com
https://assets.falcon.us-2.crowdstrike.com
https://assets-public.falcon.us-2.crowdstrike.com
https://api.us-2.crowdstrike.com
https://firehose.us-2.crowdstrike.com

# Installation and configuration of CrowdStrike Helm Charts for Kubernetes:

## Prerequisites (Preferred installation):

- Helm installed
- A machine with a container runtime running (Docker, CRI-O or Podman)
- Access to the Kubernetes API with permissions to deploy on targeted cluster (Kubectl access)

## Components Diagram:



There are 3 components to deploy:

- **The Falcon sensor (Daemonset)**: Falcon agent that provides Threat Detection and Incident and Response
- **The Falcon Kubernetes Admission Controller (Deployment)**: Falcon KAC allows for the discovery of Kubernetes objects by the Falcon Platform and evaluates Kubernetes objects against the admission control policy to identify IOMs. The admission control policy tells the Falcon KAC how to respond to new objects
- **The Falcon Kubernetes Image Assessment at Runtime or IAR (Deployment)**: Falcon IAR scans for vulnerabilities, malware, secrets and misconfigurations of container images at runtime.

## Installation Using Helm Charts

### Falcon Helm chart installation workflow

To use the Helm chart to deploy a sensor, these are the steps to follow:

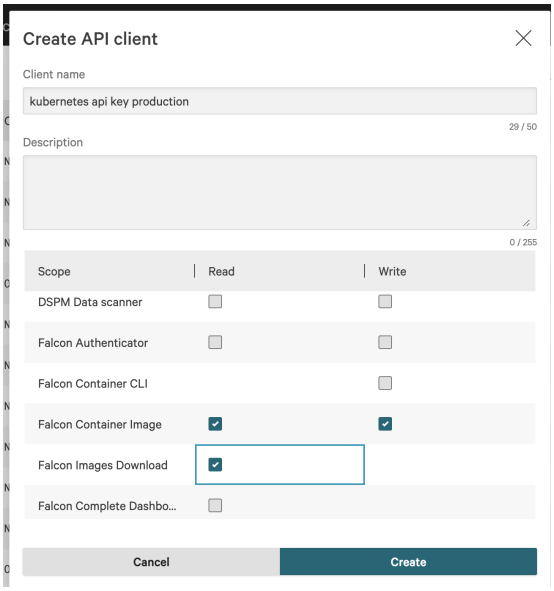
1. **Retrieve the Falcon sensor image** for your deployment
   The sensor image is available from the CrowdStrike registry. In this guide we explain how to deploy the image directly from the CrowdStrike registry, but ideally you will want to copy the image to your private registry.

2. **Set up the Helm client**
   You have two options:
   - Option 1: The recommended option is to add the Helm chart to your Helm client. To do this, you must be able to access Github on your network.
   - Option 2: If you are unable to access the Falcon Helm chart Github repo, you can download the Helm chart and run it locally. **This option is not covered on this guide**
3. **Install the sensor on your cluster**

## Configuring API Key

You can either create a single key for all components or an individual key for each component, to create an API key  on your Crowdstrike Falcon Platform connect to https://falcon.eu-1.crowdstrike.com/api-clients-and-keys/clients and click on Create API client

below here all the scopes needed for a single key, on each section the scopes needed for each component. Make sure you save the client ID and client Secret that you will be using on the following steps.




**Single API all scopes needed:**

- Falcon Image Download (read)

- Sensor Download (read)

- Falcon Container Image (read/write)

- Falcon Container CLI (write)

## FALCON SENSOR installation:

Download the script that will collect all information from the CrowdStrike API:

**API Scopes needed for Sensor only:**
- Falcon Image Download (read)
- Sensor Download (read)

**Variables (Your CrowdStrike API Credentials):**

```
export FALCON_CLIENT_ID=YOUR_CLIENT_ID
export FALCON_CLIENT_SECRET=YOUR_CLIENT_SECRET
```

**Installation**

Make sure you customize your own custom grouping tag under **`<custom-groupping-tag>`. Use the same tag as used for KAC sensor.**

Save and execute it as *install-node-sensor.sh*

```
curl -sSL -o falcon-container-sensor-pull.sh
"https://raw.githubusercontent.com/CrowdStrike/falcon-
scripts/main/bash/containers/falcon-container-sensor-pull/falcon-
container-sensor-pull.sh"
chmod +x falcon-container-sensor-pull.sh

export FALCON_CLIENT_ID=xxxxxxxx
export FALCON_CLIENT_SECRET=xxxxxxxx

echo Deploying Falcon Sensor as Daemonset
export FALCON_CID=$( ./falcon-container-sensor-pull.sh  -t falcon-sensor
--get-cid )
export FALCON_IMAGE_FULL_PATH=$( ./falcon-container-sensor-pull.sh  -t
falcon-sensor --get-image-path )
export FALCON_IMAGE_REPO=$( echo $FALCON_IMAGE_FULL_PATH | cut -d':' -f 1
)
export FALCON_IMAGE_TAG=$( echo $FALCON_IMAGE_FULL_PATH | cut -d':' -f 2
)
export FALCON_IMAGE_PULL_TOKEN=$( ./falcon-container-sensor-pull.sh  -t
falcon-sensor --get-pull-token )

helm repo add crowdstrike https://crowdstrike.github.io/falcon-helm --
force-update
helm upgrade --install falcon-sensor crowdstrike/falcon-sensor -n falcon-
system --create-namespace \
  --set falcon.cid="$FALCON_CID" \
  --set falcon.tags="<custom-groupping-tag>" \
  --set node.image.repository="$FALCON_IMAGE_REPO" \
  --set node.image.tag="$FALCON_IMAGE_TAG" \
  --set node.image.registryConfigJSON="$FALCON_IMAGE_PULL_TOKEN" \
  --set node.backend="bpf"
```

## FALCON KAC Installation

Download the script that will collect all information from the CrowdStrike API:

**API Scopes needed for KAC only:**
- Falcon Image Download (read)
- Sensor Download (read)

**Variables (Your CrowdStrike API Credentials):**

```
export FALCON_CLIENT_ID=YOUR_CLIENT_ID
export FALCON_CLIENT_SECRET=YOUR_CLIENT_SECRET
```

Make sure you customize your own custom grouping tag under **<custom-groupping-tag>. Use the same tag as used for Falcon Sensor.**

Save and execute it as *install-kac.sh*

```
curl -sSL -o falcon-container-sensor-pull.sh
"https://raw.githubusercontent.com/CrowdStrike/falcon-
scripts/main/bash/containers/falcon-container-sensor-pull/falcon-
container-sensor-pull.sh"
chmod +x falcon-container-sensor-pull.sh

export FALCON_CLIENT_ID=xxxxxxxx
export FALCON_CLIENT_SECRET=xxxxxxxx
CLUSTER_NAME="$(kubectl config view --minify -o
jsonpath='{.contexts[].context.cluster}')"

echo Deploying Falcon Admission Controller
export FALCON_CID=$( ./falcon-container-sensor-pull.sh -t falcon-kac --
get-cid )
export FALCON_IMAGE_FULL_PATH=$( ./falcon-container-sensor-pull.sh  -t
falcon-kac --get-image-path )
export FALCON_IMAGE_REPO=$( echo $FALCON_IMAGE_FULL_PATH | cut -d':' -f 1
)
export FALCON_IMAGE_TAG=$( echo $FALCON_IMAGE_FULL_PATH | cut -d':' -f 2
)
export FALCON_IMAGE_PULL_TOKEN=$( ./falcon-container-sensor-pull.sh -t
falcon-kac --get-pull-token )

helm repo add crowdstrike https://crowdstrike.github.io/falcon-helm --
force-update
helm upgrade --install falcon-kac crowdstrike/falcon-kac -n falcon-kac --
create-namespace \
  --set falcon.cid="$FALCON_CID" \
  --set falcon.tags="<custom-groupping-tag>" \
  --set image.repository="$FALCON_IMAGE_REPO" \
  --set image.tag="$FALCON_IMAGE_TAG" \
  --set clusterName="$CLUSTER_NAME" \
  --set image.registryConfigJSON="$FALCON_IMAGE_PULL_TOKEN"
```

## FALCON IAR installation

Download the script that will collect all information from the CrowdStrike API:

**API Scopes needed for IAR only:**
- Falcon Image Download (read)
- Sensor Download (read)
- Falcon Container Image (read/write)
- Falcon Container CLI (write)

**Variables (Your CrowdStrike API Credentials):**

```
export FALCON_CLIENT_ID=YOUR_CLIENT_ID
export FALCON_CLIENT_SECRET=YOUR_CLIENT_SECRET
```

**Adapt the Cloud region to your target environment:**

```
FALCON_CLOUD=us-2
FALCON_CLOUD=us-1
FALCON_CLOUD=us-gov-1
FALCON_CLOUD=us-gov-2
```

## Installation

Save and execute it as *install-iar.sh*

```
curl -sSL -o falcon-container-sensor-pull.sh
"https://raw.githubusercontent.com/CrowdStrike/falcon-
scripts/main/bash/containers/falcon-container-sensor-pull/falcon-
container-sensor-pull.sh"
chmod +x falcon-container-sensor-pull.sh

CLUSTER_NAME="$(kubectl config view --minify -o
jsonpath='{.contexts[].context.cluster}')"

export FALCON_CLIENT_ID=xxxxxxxx
export FALCON_CLIENT_SECRET=xxxxxxxx

echo Deploying Image Assessment at Runtime
export FALCON_CID=$( ./falcon-container-sensor-pull.sh -t falcon-
imageanalyzer --get-cid )
export FALCON_IMAGE_FULL_PATH=$( ./falcon-container-sensor-pull.sh -t
falcon-imageanalyzer --get-image-path )
export FALCON_IMAGE_REPO=$( echo $FALCON_IMAGE_FULL_PATH | cut -d':' -f 1
)
export FALCON_IMAGE_TAG=$( echo $FALCON_IMAGE_FULL_PATH | cut -d':' -f 2
)
export FALCON_IMAGE_PULL_TOKEN=$( ./falcon-container-sensor-pull.sh  -t
falcon-imageanalyzer --get-pull-token )

helm repo add crowdstrike https://crowdstrike.github.io/falcon-helm --
force-update
helm upgrade --install iar crowdstrike/falcon-image-analyzer -n falcon-
image-analyzer --create-namespace \
  --set deployment.enabled=true \
  --set crowdstrikeConfig.cid="$FALCON_CID" \
  --set crowdstrikeConfig.clusterName="$CLUSTER_NAME" \
  --set crowdstrikeConfig.clientID=$FALCON_CLIENT_ID \
  --set crowdstrikeConfig.clientSecret=$FALCON_CLIENT_SECRET \
  --set crowdstrikeConfig.agentRegion=$FALCON_CLOUD \
  --set image.registryConfigJSON=$FALCON_IMAGE_PULL_TOKEN \
  --set image.repository="$FALCON_IMAGE_REPO" \
  --set image.tag="$FALCON_IMAGE_TAG"
```

## All-in-One installation

If you want to deploy all 3 components at once, you can use the script below and this repository as references.

Download the script that will collect all information from the CrowdStrike API:

**API Scopes needed for all modules:**
- Falcon Image Download (read)
- Sensor Download (read)
- Falcon Container Image (read/write)
- Falcon Container CLI (write)

**Variables (Your CrowdStrike API Credentials):**

```
export FALCON_CLIENT_ID=YOUR_CLIENT_ID
export FALCON_CLIENT_SECRET=YOUR_CLIENT_SECRET
```

**Adapt the Cloud region to your target environment:**

```
FALCON_CLOUD=us-2
FALCON_CLOUD=us-1
FALCON_CLOUD=us-gov-1
FALCON_CLOUD=us-gov-2
```

### Installation

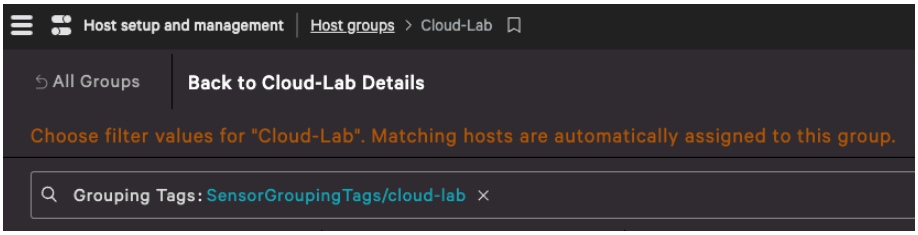
Save and execute it as *all-in-one.sh*

```
curl -O https://raw.githubusercontent.com/pbaumbach2/falcon-k8s-cluster-deploy/main/falcon-k8s-cluster-deploy.sh
chmod +x falcon-k8s-cluster-deploy.sh

export FALCON_CLIENT_ID=xxxxxxxx
export FALCON_CLIENT_SECRET=xxxxxxx
export CLUSTER_NAME="$(kubectl config view --minify -o jsonpath='{.contexts[].context.cluster}')"

cat << EOF
================
Cluster Name: $CLUSTER_NAME
EOF

./falcon-k8s-cluster-deploy.sh \
--client-id "$FALCON_CLIENT_ID" \
--client-secret "$FALCON_CLIENT_SECRET" \
--cluster "$CLUSTER_NAME" \
--region "$FALCON_CLOUD " \
--tags "pov-demo-container" \
--ebpf "true"
```

## Example of recommended detection Policy

After installation we recommend you to start with a detection only policy and leverage your previous defined grouping tag to assign a dedicated host group:

- Create a dynamic Group https://falcon.crowdstrike.com/hosts/groups-new
  - based on the tag: **`SensorGroupingTags/<custom-groupping-tag>`**




- Create a Linux Prevention policy with this config: https://falcon.crowdstrike.com/configuration/prevention/policies
- Initial Settings (For Detect Mode Only):
  - Sensor capabilities:
    - Unknown executable analysis: Enabled
    - Unknown detection-related executable analysis: Enabled
    - Sensor tamper prevention: Disabled
  - Sensor Visibility / Enhanced Visibility:
    - Enable All
  - Next-gen antivirus | On write script file visibility
    - On write script file visibility: Enabled
  - Next-gen antivirus | Quarantine
    - Quarantine: Disabled
  - Malware protection | Execution blocking
    - Custom indicator blocking: Disabled
    - Suspicious process prevention: Disabled
  - Container protection | Execution blocking
    - Container drift prevention: Disabled
  - Cloud-based anti-malware
    - Detection: Extra Aggressive
    - Prevention: Disable
  - Sensor-based anti-malware
    - Detection: Extra Aggressive
    - Prevention: Disable

Settings | Assigned host groups | Assigned custom IOAs | Audit log

Setting Name | * Required setting is O

Sensor capabilities

- Unknown executable analysis [x]
- Unknown detection-related executable analysis [x]
- Sensor tamper prevention [ ]

Sensor visibility | Enhanced visibility

- Script-based execution visibility [x]
- Filesystem visibility [x]
- Network visibility [x]
- HTTP visibility [x]
- FTP visibility [x]
- TLS visibility [x]
- Email protocol visibility [x]

Next-gen antivirus | On write script file visibility

- On write script file visibility [x]

Next-gen antivirus | Quarantine

- Quarantine [ ]

Malware protection | Execution blocking

- Custom indicator blocking [ ]
- Suspicious process prevention [ ]

Container protection | Execution blocking

- Container drift prevention [ ]

Next-gen antivirus | Cloud machine learning

Cloud-based anti-malware

| | DISABLED | CAUTIOUS | MODERATE | AGGRESSIVE | EXTRA AGGRESSIVE |
|---|---|---|---|---|---|
| Detection | | ✓ | ✓ | ✓ | ✓ |
| Prevention | ✓ | | | | |

Next-gen antivirus | Sensor machine learning

Sensor-based anti-malware

| | DISABLED | CAUTIOUS | MODERATE | AGGRESSIVE | EXTRA AGGRESSIVE |
|---|---|---|---|---|---|
| Detection | | ✓ | ✓ | ✓ | ✓ |
| Prevention | ✓ | | | | |

## Documentation

Deployment with helm charts documentation: https://github.com/CrowdStrike/falcon-helm

Deploying the Falcon sensor with helm chart: https://github.com/CrowdStrike/falcon-helm/tree/main/helm-charts/falcon-sensor

Deploying the Falcon Kubernetes Admission Controller with helm chart: https://github.com/CrowdStrike/falcon-helm/blob/main/helm-charts/falcon-kac/README.md

Deploying the Falcon Kubernetes Image Assessment at Runtime with a Helm chart: https://github.com/CrowdStrike/falcon-helm/blob/main/helm-charts/falcon-image-analyzer/README.md

Deployment with Kubernetes Operator: https://github.com/CrowdStrike/falcon-operator